

Lecture 5 & 6

Dr. Kiran Kumari



Department of Electrical Engineering
Indian Institute of Science
Bengaluru, India

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Sliding mode control and its application

Supremum and Infimum of set

Supremum of set:

Let S be a bounded subset of $\mathbb{R}$. A number ' b ' is called the least upper bound of ' S ', denoted by $b = \text{Sup}(S)$ if ' b ' satisfies the following conditions:

- ① ' b ' is an upper bound for the set.
- ② No number less than ' b ' is an upper bound for S .

Infimum of a set:

Let S be a bounded set in $\mathbb{R}$. A number ' a ' is said to be the greatest lower bound for the set ' S ', denoted by $\text{inf}(S)$ if ' a ' satisfies the following conditions:

- ① ' a ' is the lower bound of ' S '.
- ② No number greater than a is a lower bound for S .

Continuous function

Definition of continuity

Let $f : \mathbb{R} \rightarrow \mathbb{R}$. f is said to be continuous at $x \in \mathbb{R}$, if given $\epsilon > 0$, there exists $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \epsilon$$

If $f(\cdot)$ is continuous at every point in $\mathbb{R}$, then ' f ' is said to be continuous over $\mathbb{R}$.

For $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$\|x - y\| < \delta \implies \|f(x) - f(y)\| \leq \epsilon$$

Lipschitz function

Lipschitz function

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called locally Lipschitz at $x_0 \in \mathbb{R}$ if there exist ($\exists$)

- a neighbourhood $B(x_0, \epsilon)$ with $\epsilon > 0$ and
- an $L > 0$

such that

$$\|f(x_1) - f(x_2)\| \leq L\|x_1 - x_2\| \quad \text{for all } (\forall) \quad x_1, x_2 \in B(x_0, \epsilon)$$

where L is called a Lipschitz constant. Once an L is found, anything larger also works.

Note

- Any function with a bounded first derivative must be a Lipschitz function.
- x^2 , e^x and e^{-x} are few examples of functions which are locally Lipschitz on $\mathbb{R}$ but not globally Lipschitz.
- $\sin x$, $\cos x$, ax where a is constant are globally Lipschitz function.

Existence and uniqueness of solution

Existence and uniqueness for $\dot{x} = A(t)x + g(t)$:

Suppose that $A(t)$ and $g(t)$ are continuous on some open interval $J \subset \mathbb{R}$. Then, for any $t_0 \in J$ and any $x_0 \in \mathbb{R}^n$, the above equation has a unique solution satisfying $x(t_0) = x_0$.

Piece wise continuous function

There is a finite number of breaks in the function, and it does not blow to infinity.

Local existence and uniqueness for $\dot{x} = f(t, x)$:

Let $f(t, x)$ be piecewise continuous in t and satisfy the Lipschitz condition

$$\|f(t, x) - f(t, y)\| \leq L\|x - y\|$$

$\forall x, y \in B = \{x \in \mathbb{R}^n : \|x - x_0\| \leq r\} \forall t \in [t_0, t_1]$. Then, there exists some $\delta > 0$ such that the state equation $\dot{x} = f(t, x)$ with $x(t_0) = x_0$ has a unique solution over $[t_0, t_0 + \delta]$.

For global existence and uniqueness,

$$\|f(t, x) - f(t, y)\| \leq L\|x - y\| \quad \forall x, y \in \mathbb{R}^n, \forall t \in [t_0, t_1].$$

Stability of origin

Any equilibrium point can be shifted to the origin via a change of variables.

The equilibrium point $x = 0$ of $\dot{x} = f(x)$ is

- stable if, for each $\epsilon > 0$, there is $\delta = \delta(\epsilon) > 0$ such that

$$\|x(0)\| < \delta \implies \|x(t)\| < \epsilon, \forall t \geq 0$$

- unstable if it is not stable.
- asymptotically stable if it is stable and $\delta > 0$ can be chosen such that

$$\|x(0)\| < \delta \implies \lim_{t \rightarrow \infty} x(t) = 0$$

Lyapunov's Stability theorem

Theorem 4.1 from Khalil

Let $x = 0$ be an equilibrium point for $\dot{x} = f(x)$ and $D \subset \mathbb{R}^n$ be a domain containing $x = 0$. Let $V : D \rightarrow \mathbb{R}$ be a continuously differentiable function such that

$$V(0) = 0 \quad \text{and} \quad V(x) > 0 \text{ in } D - \{0\} \quad (1)$$

$$\dot{V}(x) \leq 0 \text{ in } D. \quad (2)$$

Then, $x = 0$ is stable.

Moreover if

$$\dot{V}(x) < 0 \text{ in } D - \{0\} \quad (3)$$

then $x = 0$ is asymptotically stable.

- A function $V(x)$ satisfying condition (1), that is, $V(0) = 0$ and $V(x) > 0$ for $x \neq 0$ is said to be positive definite.
- If satisfies weaker condition $V(x) \geq 0$ for $x \neq 0$, it is said to be positive semidefinite.
- $V(x)$ is negative definite or negative semidefinite if $-V(x)$ is positive definite or positive semidefinite, respectively.

Lyapunov's Stability theorem

Lyapunov candidate function

A real symmetric matrix P is called positive definitive matrix if $x^T P x > 0 \forall x \neq 0$, and $x^T P x = 0$ for $x = 0$, where $x \in \mathbb{R}^n$.

The quadratic function $V = x^T P x$ is called the quadratic Lyapunov function.

Theorem for global stability

Let $x = 0$ be an equilibrium point. Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuously differentiable function such that,

$$V(0) = 0 \quad \text{and} \quad V(x) > 0 \forall x \neq 0 \quad (4)$$

$$\|x\| \rightarrow \infty \implies V(x) \rightarrow \infty \quad (5)$$

$$\dot{V}(x) < 0 \quad \forall x \neq 0 \quad (6)$$

then, $x = 0$ is globally asymptotically stable.

The function satisfying condition (5) is said to be radially unbounded.

Hurwitz matrix

A matrix A is Hurwitz; that is, $\text{Re}\lambda_i < 0$ for all eigenvalues of A , if and only if for any given positive definite symmetric matrix Q there exists a positive definite symmetric matrix P that satisfies the Lyapunov equation

$$A^T P + PA = -Q. \quad (7)$$

Moreover, if A is Hurwitz, then P is the unique solution of (7).

The Invariance Principle

Invariant set

A set M is said to be invariant with respect to $\dot{x} = f(t, x)$ if $x(0) \in M \implies x(t) \in M$, $\forall t \in \mathbb{R}$.

This means if a solution belongs to M at some time instant, then it belongs to M for all future and past time.

Positively Invariant set

A set M is said to be positively invariant with respect to $\dot{x} = f(t, x)$ if $x(0) \in M \implies x(t) \in M$, $\forall t \geq 0$

LaSalle's theorem

Let $\Omega \subset D$ be a compact set that is positively invariant with respect to $\dot{x} = f(t, x)$. Let $V : D \rightarrow \mathbb{R}$ be a continuously differentiable function such that $\dot{V}(x) \leq 0$ in Ω . Let E be the set of all points in Ω where $\dot{V}(x) = 0$. Let M be the largest invariant set in E . Then, every solution starting in Ω approaches M as $t \rightarrow \infty$.

Corollary

Let $x = 0$ be an equilibrium point for $\dot{x} = f(t, x)$. Let $V : D \rightarrow \mathbb{R}$ be a continuously differentiable positive definite function on a domain D containing the origin $x = 0$, such that $\dot{V}(x) \leq 0$. Let $S = \{x \in D \mid \dot{V}(x) = 0\}$ and suppose that no solution can stay identically in S , other than the trivial solution $x(t) = 0$. Then, the origin is asymptotically stable.

Rayleigh inequality

$$\lambda_{\min}(P)\|x\|^2 \leq x^T P x \leq \lambda_{\max}(P)\|x\|^2$$

Comparison Principle

Comparison lemma

Consider the scalar differential equation

$$\dot{u} = f(t, u), \quad u(t_0) = u_0$$

where $f(t, u)$ is continuous in t and locally Lipschitz in u , for all $t \geq 0$ and all $u \in J \subset \mathbb{R}$. Let $[t_0, T)$ (T could be infinity) be the maximal interval of existence of the solution $u(t)$, and suppose $u(t) \in J$ for all $t \in [t_0, T)$.

Let $v(t)$ be a continuous function whose upper right-hand derivative satisfies the differential inequality

$$D^+v(t) \leq f(t, v(t)), \quad v(t_0) \leq u_0$$

with $v(t) \in J \forall t \in [t_0, T)$. Then, $v(t) \leq u(t)$ for all $t \in [t_0, T)$.

Upper right-hand derivative

$$D^+v(t) = \lim_{h \rightarrow 0^+} \sup \frac{v(t+h) - v(t)}{h}.$$

THANK YOU